

Contact

+32 439 76 12 99 (Mobile)
lena.vermeulen@example.com
www.linkedin.com/in/lenavermeulen
(LinkedIn)
lenavermeulen.example.com
(Personal)

Top Skills

Python
PyTorch
LangChain
RAG
Kubernetes

Languages

Dutch (Native)
English (Fluent)
French (Professional working proficiency)

Lena Vermeulen

AI Engineer | Building LLM-powered systems for industry | ex-data engineer
Antwerp, Flanders, Belgium

Summary

AI engineer with 10 years of experience spanning data engineering and machine learning. I design and deploy LLM-based solutions that turn unstructured data into actionable insights for manufacturing and logistics. Known for bridging the gap between research and production, with a focus on reliability and measurable business impact.

Experience

Cortexa AI

Lead AI Engineer, LLM Systems
August 2024 - Present (2 years 2 months)
Antwerp, Belgium

Currently building production-grade LLM applications for document understanding and conversational analytics in the manufacturing sector.

- Developed a retrieval-augmented generation (RAG) pipeline that reduced document search time by 70% for engineers.
- Fine-tuned a domain-specific LLM on technical manuals, achieving 92% accuracy on Q&A benchmarks.
- Led the integration of LLM outputs into existing BI dashboards, enabling natural language queries for non-technical users.

Volta Energy

Senior AI Engineer
December 2022 - July 2024 (1 year 8 months)
Brussels, Belgium

Led AI initiatives for a renewable energy company, focusing on forecasting and optimization of energy grids.

- Designed a time-series forecasting model that improved wind energy prediction accuracy by 15%.
- Implemented a reinforcement learning agent for battery storage optimization, increasing revenue by 12%.
- Mentored a team of 3 junior engineers and established MLOps best practices.

Nexa Robotics

AI Engineer

June 2020 - November 2022 (2 years 6 months)

Ghent, Belgium

Joined a robotics spin-off to develop computer vision and NLP capabilities for autonomous warehouse robots.

- Built a real-time object detection system using YOLOv5 that improved picking accuracy by 22%.
- Created a natural language interface for robot commands, enabling non-technical staff to operate robots.
- Optimized model inference on edge devices, reducing latency from 200ms to 50ms.

Dataflow Analytics

4 years 3 months

Machine Learning Engineer

September 2018 - May 2020 (1 year 9 months)

Antwerp, Belgium

Transitioned to building ML models for predictive maintenance and demand forecasting, taking ownership of the full ML lifecycle.

- Developed a predictive maintenance model that reduced unplanned downtime by 18% for a major client.
- Implemented a feature store that accelerated model development by 50%.
- Led the deployment of real-time inference endpoints using Kubernetes, handling 10k requests per second.

Data Engineer

March 2016 - August 2018 (2 years 6 months)

Antwerp, Belgium

Built and maintained data pipelines for a logistics analytics platform, ensuring reliable data delivery for downstream reporting.

- Reduced pipeline runtime by 40% by optimizing Spark jobs and introducing incremental processing.
- Designed a data quality monitoring system that cut data errors by 25%.
- Migrated on-premise Hadoop cluster to AWS EMR, saving 30% in infrastructure costs.

Education

KU Leuven

Master of Science, Computer Science · (2011 - 2015)